

3	(a) acceleration/force proportional to displacement (from fixed point)	M1	
	acceleration/force and displacement in opposite directions	A1	[2]
	(b) <u>maximum</u> displacements/accelerations are different	B1	
	graph is curved/not a straight line	B1	[2]
	(c) (i) $\omega = 2\pi / T$ and $T = 0.8\text{ s}$	C1	
	$\omega = 7.9\text{ rad s}^{-1}$	A1	[2]
	(ii) $a = (-)\omega^2 x$ $= 7.85^2 \times 1.5 \times 10^{-2}$	C1	
	$= 0.93\text{ ms}^{-2}$ or 0.94 ms^{-2}	A1	[2]
	(iii) $\Delta E = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$	C1	
	$= \frac{1}{2} \times 120 \times 10^{-3} \times 7.85^2 \times \{(1.5 \times 10^{-2})^2 - (0.9 \times 10^{-2})^2\}$	C1	
	$= 5.3 \times 10^{-4}\text{ J}$	A1	[3]